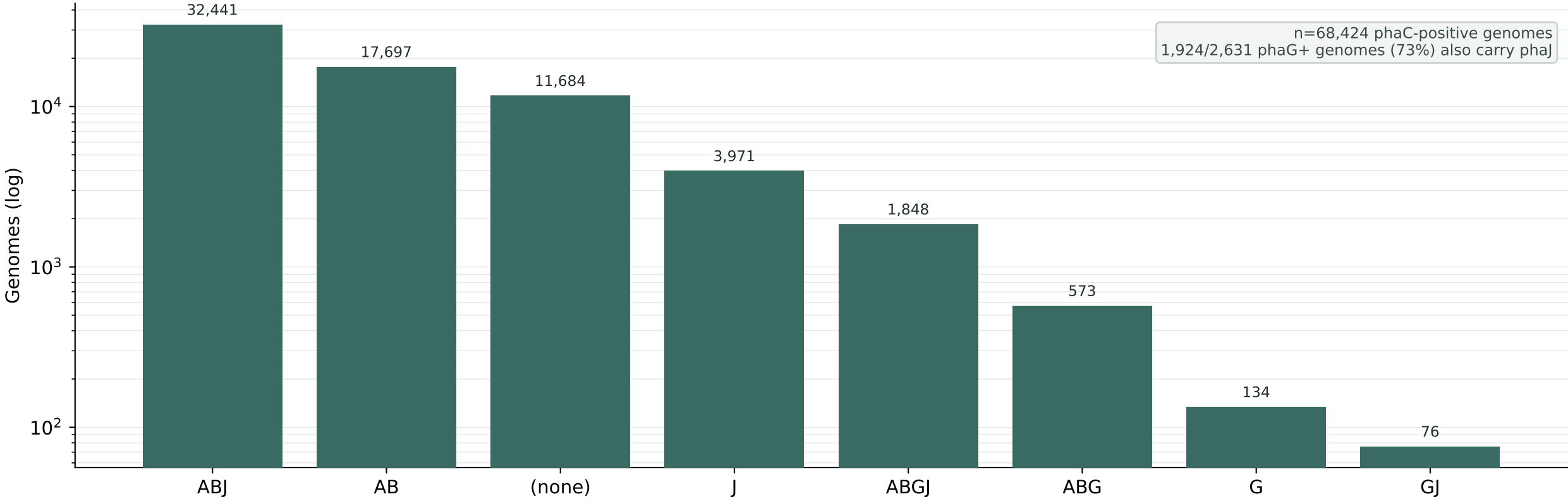


mcl-PHA precursor-supply strategy: scavenging (phaJ) vs. de novo synthesis (phaG)

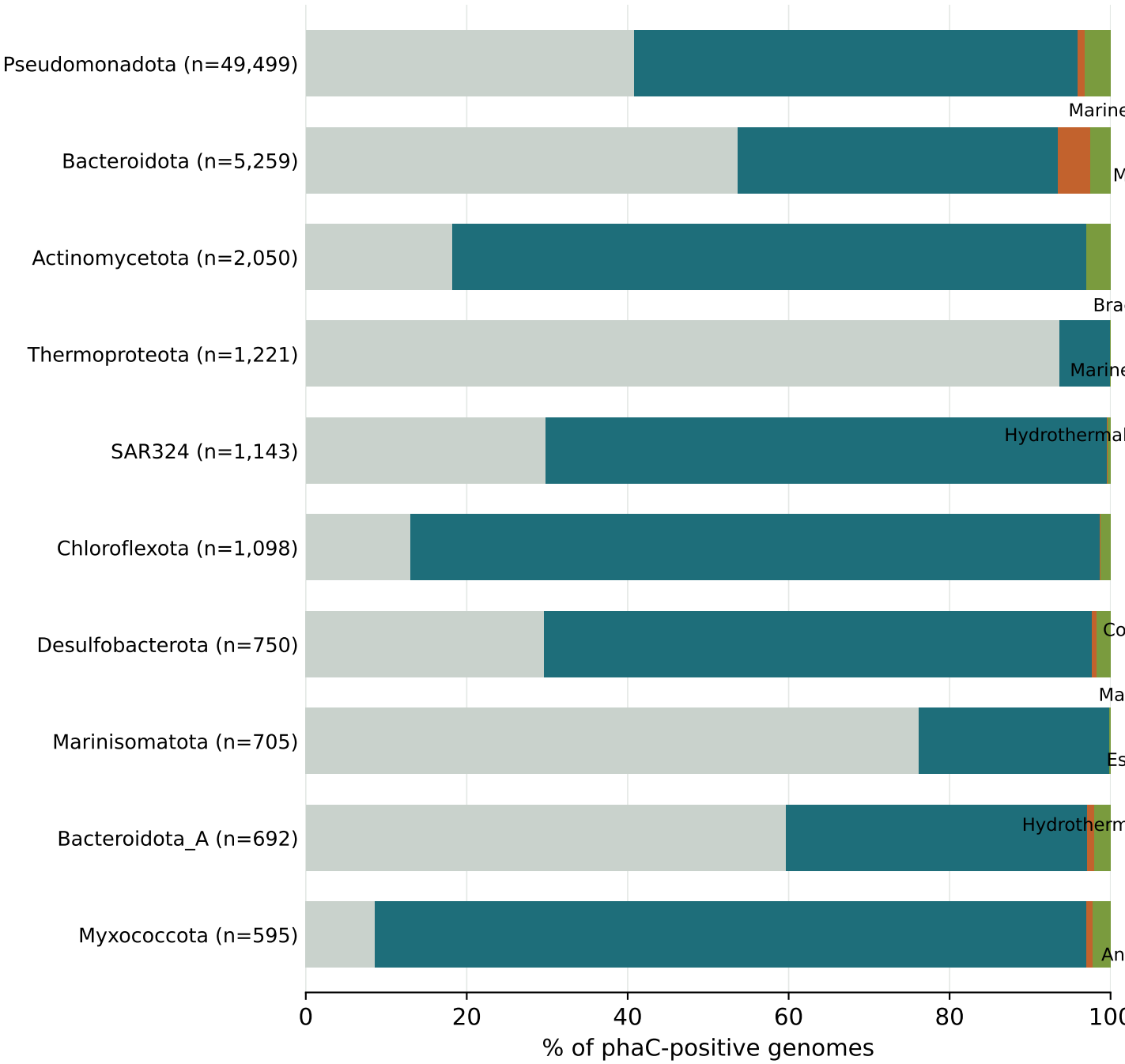
phaC-positive genomes only (n=68,424). "De novo" and "Both" bars are phaG-positive; phaG is rare overall (3.8% of phaC-positive genomes) and mostly co-occurs with phaJ rather than replacing it.

A. Full pathway architecture: canonical (AB) x scavenging (J) x de novo (G)

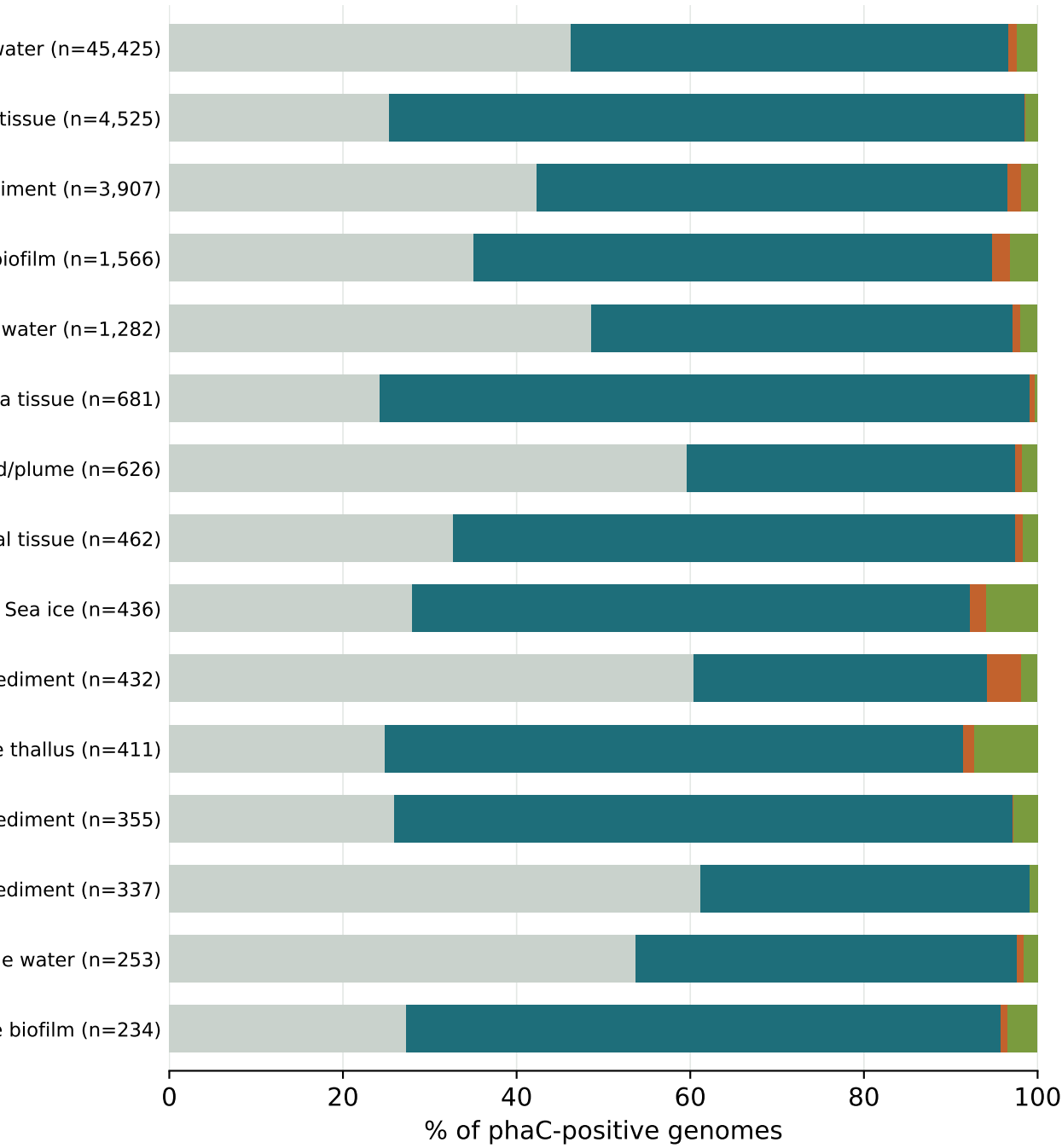


n=68,424 phaC-positive genomes
1,924/2,631 phaG+ genomes (73%) also carry phaJ

B. MCL precursor strategy by phylum (top 10)



C. MCL precursor strategy by habitat



Neither Scavenging only (J) De novo only (G) Both (G+J)